

Riley C. Yeoman

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EDUCATION

University of Minnesota – Twin Cities

Master of Science in Robotics

Minneapolis, MN

Dec. 2024

University of Oklahoma

Bachelor of Science in Biomedical Engineering

Norman, OK

May 16, 2021

SKILLS & TOOLS

Skills: ROS, Linux/Unix, Machine Learning, DirectX, Git, Computer Vision, OpenGL, Keras, and PyTorch.

Programming Languages: C++, C, Python, C#, and MATLAB

RESEARCH EXPERIENCE

MIND

University of Minnesota

Masters Research

Sept 2023 – Current

- Developed MIND (Mental Health Identification with No-label Distillation), a machine learning model utilizing transformer architecture alongside self-supervision for computer vision tasks.
- Collaborated with experienced professors and professionals to explore and apply current machine learning techniques in robotics.
- Leveraged MIND's capabilities to identify mental health states, including anxiety and depression, with potential to detect additional conditions in the future.

WORK EXPERIENCE

Medtronic

Mounds View, MN

Reliability Engineer

Jan 2022 – Dec 2022

- Remodeled and upgraded outdated software systems in catheter testing equipment programmed in Python and C++ to automate equipment testing functions.
- Provided sole technical expertise in software, CAD design, and electronics for the entire Cardiac Ablation Solutions (CAS) Post-Market Reliability team, designing physical fixturing and writing scripting software for custom equipment.
- Authored experimental protocols to test requirements for products under the electrophysiology (EP) product scope after design changes to ensure repeatable functionality under ISO and FDA requirements and regulations.
- Designed and implemented specialized vision systems for testing equipment, assuming sole responsibility for their development and maintenance. This expertise ensured the seamless integration of vision technology into the testing process, guaranteeing precise and efficient assessments while minimizing operational disruptions.

PROJECT EXPERIENCE

Acute Catheter Actuation Testing (ACAT)

Medtronic

Software Design/Reliability Engineer

Jan 2022-Dec 2022

- Developed a vision measurement system that harnessed machine learning and computer vision methods to recognize catheters in real-time medical imagery, facilitating precise tracking and measurement of their movement distance and speed during procedures such that they follow FDA regulations.

CSGO-Trajectory Generation

UMN

Group Project

2024

- Leveraged open-source Counter Strike Global Offensive player positioning data to develop simulations of agent movement, utilizing transformers and reinforcement learning to model and optimize behavior patterns.

Ray Tracer

Open-Source

Personal Project

2024

- Designed and implemented a custom ray tracer from scratch in C++, simulating complex light interactions with 3D objects, including accurate modeling of shadows, reflections, refractions, and surface materials.